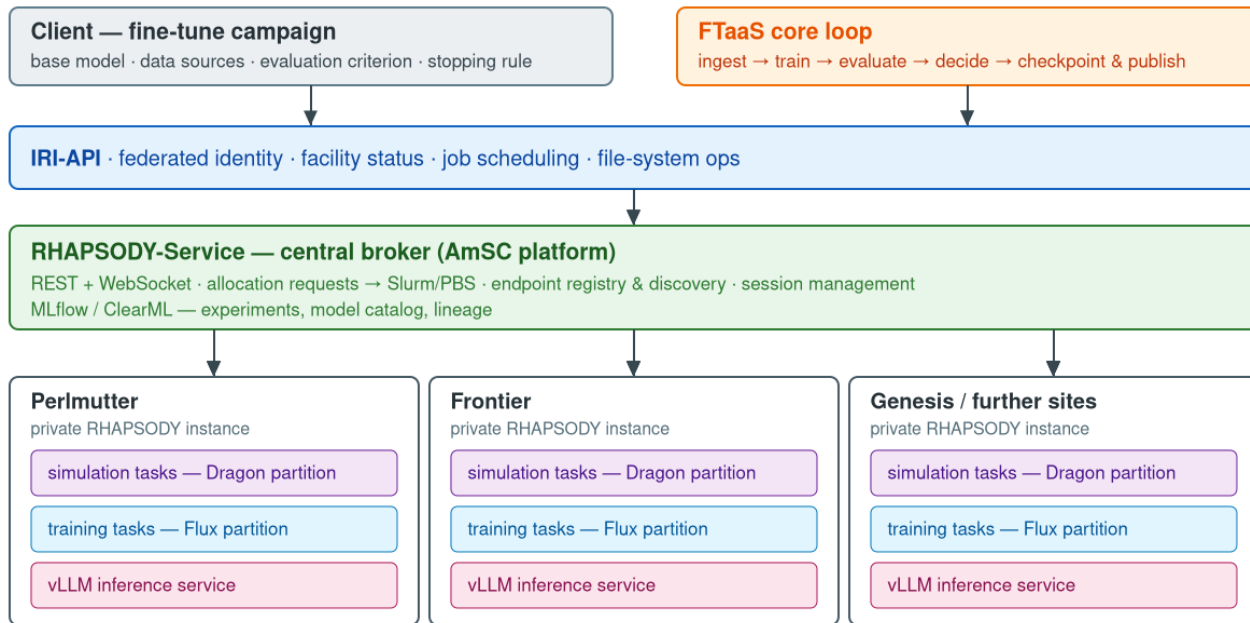


# Fine-Tuning as a Service on the AmSC — Design



One allocation per site holds simulation, training and inference side by side — endpoints dial out, no inbound firewall exceptions.

# FTaaS — where we stand

## **Runs on Perlmutter today**

- Fine-tuning campaign (MATEY) driven through the IRI / SFAPI path.
- Simulation and training side by side in one allocation — concurrent Dragon and Flux partitions, validated against Flux 0.79.
- Multiple inference services per allocation, explicit node placement, typed model / hardware / batching configuration.

## **RHAPSODY at scale — the substrate this rests on** (arXiv 2512.20795)

- ~11k tasks/s sustained peak, near-linear weak scaling, a distinct task on every node.
- ~50,000 active agents: sustained overlap of AI decision rate and HPC task realization.
- 100–300  $\mu$ s per-task overhead, 90–95% GPU utilization; Frontier, Perlmutter, Anvil.

**Open work** — the submission surface that turns the recipe into a service; concurrent fine-tunes sharing one allocation; catalog, lineage and promotion to downstream tasks.